

What Can Three Seeds and Fifty Rollouts Resolve? A Preregistered Audit of Operator and Seed Variance on robomimic Multi-Human Benchmarks

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Abstract—Results on the robomimic Multi-Human (MH) imitation-learning benchmarks are conventionally reported as a mean over three training seeds, each evaluated with 50 rollouts. We preregistered the hypothesis that the variance caused by *which operators’ demonstrations are in the training set* exceeds the variance caused by training seed, and tested it with 138 BC-RNN runs on Lift, Can and Square: ten seeds at fixed data, six leave-one-operator-out partitions, and six size-matched random partitions as a control, all evaluated at the final checkpoint on fixed initial states. The preregistered tests were not met on any task; with six operators the procedure detects a true ratio of two only 18% of the time. What the data do establish concerns the reporting convention. With 50 rollouts on shared initial states, checkpoints trained with different seeds cannot be told apart by success rate on any task; they differ mainly in which states they solve. On Square, two three-seed means must differ by 12.7 points to be distinguished. Re-evaluating the same 46 Square checkpoints with 500 rollouts, the seed effect becomes detectable, the 50-rollout success rates correlate only $r = 0.56$ with the 500-rollout ones, and an apparent proficiency-tier pattern disappears. At 500 rollouts the operator-partition standard deviation is 0.054 against 0.014 for random partitions; that contrast nominally meets the preregistered threshold but does not survive Monte Carlo error, and we do not treat it as established. We give reporting recommendations.

Index Terms—Imitation learning, benchmarking, reproducibility, variance components, learning from demonstration.

I. INTRODUCTION

THE robomimic study [1] established the most widely used protocol for evaluating offline imitation learning in manipulation: train on a fixed demonstration dataset, evaluate with 50 rollouts, and report success rate averaged over three seeds. Its Multi-Human (MH) datasets, 300 demonstrations per task from six operators of varied proficiency collected with RoboTurk [2], are a standard testbed, and later work reports on them with the same three seeds and 50 evaluation episodes, for example Diffusion Policy and its baselines [3]. The three-seed standard deviation is the only uncertainty attached to such numbers.

That standard deviation measures how much the result moves when the optimizer’s random stream changes and everything else is fixed. It says nothing about how much the result would move had the dataset contained demonstrations

from different people. For a benchmark built from six operators, that second source is not obviously smaller. If it is larger, the conventional uncertainty understates the real one.

We preregistered a test of that claim before any training run. The claim was not supported, and we report that plainly, together with what the experiment did establish, which concerns the measurement rather than the operators:

- **A preregistered negative result with its power** (Section IV-B). With six operators the procedure has 18% power at a true ratio of two, so the negative result bounds little.
- **At 50 rollouts the seed effect is invisible** (Section IV-A). An exact paired test finds no difference between ten seeds on any task; checkpoints differ in which states they solve, and that only averages out over many states.
- **Fifty rollouts do not characterise a checkpoint** (Section IV-C). At 500 rollouts the seed effect on Square is detectable, the 50-rollout numbers explain 31% of the variance of the 500-rollout ones, and a tier pattern seen at 50 rollouts vanishes.
- **Resolvability thresholds and recommendations** (Sections IV-E and V).

We cite robomimic as the origin of a convention the field, including this author, has used; nothing here criticises that study, which also released the per-operator labels that make this audit possible.

II. RELATED WORK

Seeds and statistical practice. Seed sensitivity and the unreliability of small seed counts are well documented in deep reinforcement learning [4], [5], [6]. Bouthillier *et al.* [7] showed for supervised benchmarks that data sampling, initialisation and hyperparameters all markedly affect comparisons; our Arm C is the imitation-learning analogue of their data-resampling source. Kress-Gazit *et al.* [8] recommend reporting evaluation conditions and statistical uncertainty in robot learning. We measure how the conventional statistic decomposes on a heavily used benchmark.

Demonstrator heterogeneity. The robomimic study analysed MH data by proficiency tier, pooling two operators per tier, to ask how algorithms cope with suboptimal data [1]; individual operators were not a unit of analysis. Data quality and the compatibility of demonstrators affect imitation learning [9], [10]. Our question is statistical: with the operator as the unit of replication, how much does performance move when

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Code, preregistration, partitions, initial states, per-run and per-rollout results: <https://github.com/imjbassi/operator-variance-audit>, archived at <https://doi.org/10.5281/zenodo.23109850>.

one operator is removed, relative to seed and to removing the same number of random demonstrations? We follow preregistration practice [11], with a dated amendment log.

III. PREREGISTERED DESIGN

The preregistration, with predictions, falsification criteria and the analysis plan, was committed to the public repository before any training run and is unchanged there, followed by four dated amendments.

A. Data, training and evaluation

We use the robomimic v1.5 MH low-dimensional datasets for Lift, Can and Square (robosuite [12] 1.5.1): 300 demonstrations each, 50 per operator. Operator identity is read directly from the filter keys shipped in the files (`better_operator_1`, ..., `worse_operator_2`), which we verified are disjoint, contiguous blocks of 50 covering the dataset. The operator index is defined within a tier and is not documented to denote the same person across tasks, so we make no cross-task statement about individuals.

All runs train BC-RNN with the robomimic low-dimensional hyperparameters (two-layer LSTM of width 400, five-mode Gaussian-mixture head, sequence length 10, Adam at 10^{-4} , batch 100, 2000 epochs of 100 steps; robomimic v0.5.0). We evaluate only the *final* checkpoint: the original study reports the maximum over periodic evaluations [1], which is selection on the test set. Our final-checkpoint means on the full datasets are 0.988, 0.946 and 0.600; the maximum-over-training figures in [1, Table 1] are 100.0, 100.0 and 78.0, under a different checkpoint rule and dataset version. Each checkpoint is evaluated with 50 rollouts of horizon 500 on 50 initial conditions generated once per task, stored, and reused for every checkpoint. Both the simulator state and the scene model are restored, since robosuite regenerates the model at each reset and Lift randomises the cube size, and the rollout random stream is fixed per rollout index. Re-running a checkpoint reproduced every rollout outcome and step count.

B. Three arms

Arm A (seed): all 300 demonstrations, ten seeds. **Arm B (operator composition):** six leave-one-operator-out partitions of 250 demonstrations, three seeds each. **Arm C (matched random control):** six partitions of 250 demonstrations drawn uniformly at random from the 300, three seeds each. That is 10, 18 and 18 runs per task, 138 runs and 6,900 rollouts in total. Arm C is load-bearing: two Arm B partitions differ in which operator is absent but also in which demonstrations are present (any two share 200 of 250). Arm C has the same size and comparable overlap and isolates the second effect. Partition memberships were committed once and never changed.

C. Statistics

The unit of replication is the operator (the partition), never the rollout or the demonstration; the per-run statistic is the final-checkpoint success rate y .

Seed spread. s_A is the standard deviation over the ten Arm A seeds, with a seed-bootstrap interval, measured on the shared stored states. The binomial floor $\sqrt{\bar{p}(1-\bar{p})/n}$ is the sampling error of one success rate against a fresh draw of n states. On shared states, outcomes are correlated across checkpoints, so the null for “no difference between seeds” is the per-state floor $\sqrt{q_i(1-q_i)/n}$, with q_i the success rate on state i across checkpoints, and the exact test is Cochran’s Q on the checkpoints-by-states matrix. The per-state floor and Cochran’s Q are post hoc (Amendment 4).

Variance components. For Arms B and C separately, the 6×3 partition-by-seed table is modelled as $y_{ps} = \mu + a_p + b_s + e_{ps}$ with independent effects of variance σ_p^2 , σ_s^2 , σ_e^2 , estimated by the method of moments, $\hat{\sigma}_p^2 = (\text{MS}_p - \text{MS}_e)/3$ and $\hat{\sigma}_s^2 = (\text{MS}_s - \text{MS}_e)/6$ truncated at zero [13], and by REML [14], [15]. The residual absorbs rollout noise and partition-by-seed interaction. A zero estimate is a boundary estimate, not evidence of zero variance.

Contrasts and decision rule. The headline ratio $R = \hat{\sigma}_p^B / s_A$ has a percentile interval and one-sided p -value from a joint bootstrap [16] over the six partitions and ten Arm A seeds (2000 resamples). The control contrast $D = (\hat{\sigma}_p^B)^2 - (\hat{\sigma}_p^C)^2$, positive when operator partitions vary more than random ones, is bootstrapped over the six partitions of each arm. The primary family, $R > 1$ and $D > 0$ on three tasks, is corrected by Holm’s procedure [17]; a contrast is supported if its Holm-adjusted one-sided p is below 0.05, with intervals shown unadjusted. The preregistration stated both an interval criterion and Holm correction without saying which governs; Amendment 4 fixes this rule.

Secondary claim. Tier labels were predicted not to capture the operator effect, judged by the between-tier share of the Arm B partition-mean sum of squares against relabelling.

D. Amendments

Amendment 1, before any data: the tier permutation test has only 15 distinct outcomes ($p \geq 1/15$), so the secondary claim is judged by effect size; power is reported by simulating the headline procedure. *Amendment 2*, after Stage 1 and before any further rollout: Stage 2 re-evaluates the 46 Square final checkpoints with 500 rollouts on 500 new fixed states, with the same analysis and a two-contrast Holm family; being designed after Stage 1 and reusing its checkpoints, it is a follow-up, not independent confirmation. It also declares three exploratory analyses (Section IV-F). *Amendment 3* records the Stage 2 outcome and two reporting corrections. *Amendment 4*, after external review with all results known, fixes the decision rule above and adds the post hoc analyses labelled as such below.

IV. RESULTS

A. The seed effect at 50 rollouts

Fig. 1 shows every run and Table I the estimates. At fixed data the ten-seed standard deviation is 0.017 on Lift, 0.023 on Can and 0.079 on Square, where the seeds range over 22 points. **No task shows a detectable seed effect at 50 rollouts.** Cochran’s Q gives $p = 0.213, 0.855$ and 0.133 (Table II).

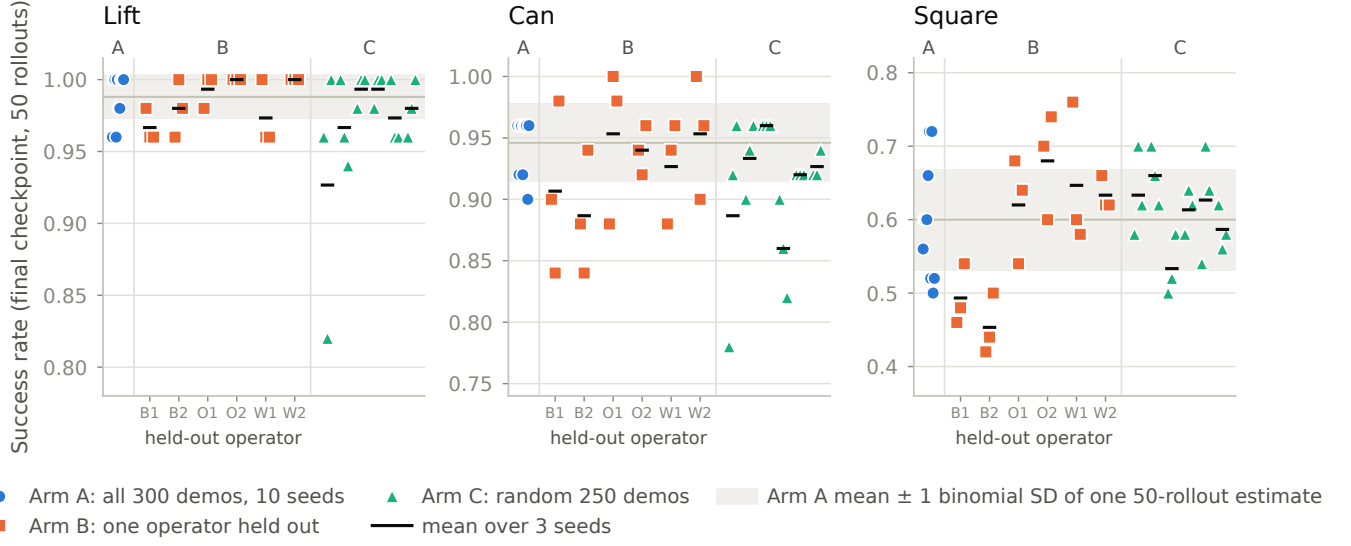


Fig. 1. Stage 1. Final-checkpoint success rate of every run (50 rollouts). A: ten seeds on all 300 demonstrations. B: leave-one-operator-out partitions, three seeds each, labelled by the held-out operator (B/O/W: better/okay/worse tier). C: random 250-demonstration partitions, three seeds each. Bars are three-seed means; the grey band is the Arm A mean $\pm$ one binomial SD of a single 50-rollout estimate. Vertical scales differ.

TABLE I

VARIANCE COMPONENTS (METHOD OF MOMENTS; REML AGREES TO WITHIN 0.003 ON EVERY ENTRY) AND THE PREREGISTERED CONTRASTS. ARM A SD WITH ITS 95% SEED-BOOTSTRAP INTERVAL; “FLOOR” IS THE BINOMIAL SD AT THE ARM A MEAN. INTERVALS ARE FROM THE PREREGISTERED 2000 RESAMPLES; FOR D , p -VALUES ARE PREREGISTERED RESAMPLING / EXACT ENUMERATION. $\dagger$ BOUNDARY ESTIMATE, NOT EVIDENCE OF ZERO VARIANCE.

Task	Arm A (10 seeds)			Arm B (operator held out)			Arm C (random 250)			R (Sec. III-C)		$D \times 10^3$ (Sec. III-C)		
	mean	SD	floor	$\hat{\sigma}_p$	$\hat{\sigma}_s$	$\hat{\sigma}_e$	$\hat{\sigma}_p$	$\hat{\sigma}_s$	$\hat{\sigma}_e$	est. [95%]	p_{Holm}	est. [95%]	p	p_{Holm}
<i>Stage 1 (preregistered primary): 50 rollouts per checkpoint</i>														
Lift	0.988	0.017 [0.006, 0.020]	0.015	0.011	0.000 †	0.015	0.000 †	0.000 †	0.043	0.66 [0.00, 1.35]	1.000	0.12 [-0.15, 0.25]	0.1880 / 0.1937	0.9400 / 0.9684
Can	0.946	0.023 [0.000, 0.029]	0.032	0.004	0.025	0.046	0.024	0.000 †	0.045	0.18 [0.00, 1.64]	1.000	-0.58 [-1.71, 0.44]	0.8010 / 0.7933	1.0000 / 1.0000
Square	0.600	0.079 [0.044, 0.096]	0.069	0.086	0.027	0.057	0.033	0.007	0.050	1.08 [0.00, 1.84]	1.000	6.22 [-1.35, 10.74]	0.0810 / 0.0844	0.4860 / 0.5062
<i>Stage 2 (follow-up, Amendment 2): same checkpoints, 500 rollouts</i>														
Square	0.639	0.031 [0.017, 0.039]	0.021	0.054	0.000 †	0.045	0.014	0.010	0.045	1.71 [0.51, 2.96]	0.115	2.69 [0.02, 5.12]	0.0245 / 0.0251	0.0490 / 0.0501

Against the binomial floor the picture is the same: the Can spread is below it, and the Lift and Square spreads exceed it by amounts a χ^2_9 test does not distinguish from zero ($p = 0.290, 0.223$). The checkpoints are not identical: on Square 46 of the 50 states are solved by some seeds and not others. Seeds differ far more in *which* states they solve than in how many, and that only averages out over many states.

B. Preregistered primary tests (Stage 1)

The primary claim is not supported on any task. $R = 1.08 [0.00, 1.84]$ on Square, 0.66 on Lift and 0.18 on Can, all with Holm $p = 1.000$. The control contrast is positive on Square (partition SD 0.086 in Arm B, 0.033 in Arm C) but not supported ($p = 0.0810$, Holm 0.4860). On Can the random partitions vary more than the operator partitions; on Lift, at ceiling, several components are at the boundary. The two estimators agree everywhere.

This negative result is weak evidence about effect size. Simulating the preregistered procedure at the fitted noise levels (Fig. 2), power on Square is 18%, 42% and 66% at true ratios of two, three and four, and no better elsewhere. Six partitions give five degrees of freedom, and a bootstrap over six clusters

is unreliable [18]. The denominator s_A also contains the rollout noise floor while the numerator is residual-corrected, so the ratio is conservative: its false-positive rate at a ratio of one is at most 0.5%, against a nominal 2.5%. We noted this when Arm A finished, recorded it before any Arm C result existed, and did not alter the test.

C. Stage 2: the same checkpoints at 500 rollouts

A 50-rollout success rate is a poor measurement of a checkpoint (Fig. 3). Across the 46 checkpoints the two evaluations correlate at $r = 0.56$. Their differences have standard deviation 0.068, matching the binomial expectation of 0.071, and reach 16 points. The 50-state set was also 3.3 points harder on average, so results on a small fixed set are conditional on it.

The seed effect becomes visible. The ten-seed standard deviation is 0.031 [0.017, 0.039], against floors of 0.021 (binomial) and 0.020 (per-state). Cochran’s $Q = 22.0$ on 9 degrees of freedom ($p = 0.009$): seeds produce checkpoints whose success rates differ with a standard deviation of about 0.024. Fifty states could not resolve this.

The operator contrast nominally meets the preregistered threshold; we do not treat it as established. The partition

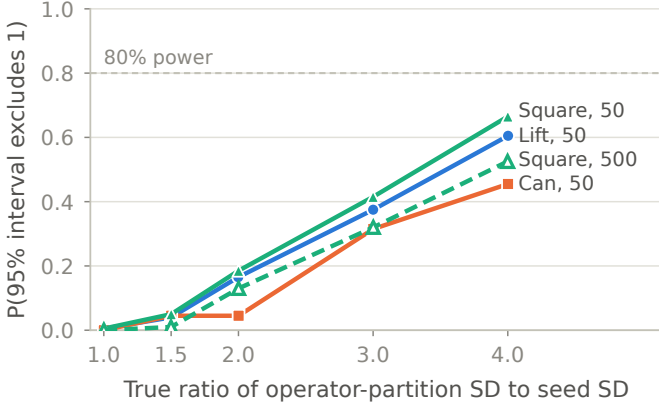


Fig. 2. Power of the preregistered headline procedure (probability that the 95% interval for R excludes one) versus the true ratio, simulated at each task’s fitted noise levels. Solid: 50 rollouts; dashed: Square at 500. Each point is 200 simulations (Monte Carlo SE up to 3.5 points).

standard deviation is 0.054 in Arm B and 0.014 in Arm C, with equal residuals (0.045). $R = 1.71$ [0.51, 2.96] is not supported ($p = 0.115$). For D , the preregistered bootstrap gives $D \times 10^3 = 2.69$ [0.02, 5.12], $p = 0.0245$, Holm 0.0490. *As preregistered, this meets the threshold*, under the rule of Section III-C and under an unadjusted interval criterion, but not robustly: the Monte Carlo standard error of the Holm p is 0.0069, well above the margin of 0.001. After seeing this we enumerated the bootstrap exactly (462 distinct resamples of six partitions), which gives $p = 0.0251$, Holm 0.0501, interval $[-0.00, 5.23]$. Not claiming the contrast is therefore a conservative choice made after the fact. Both stages put operator partitions above random ones; the data are consistent with an operator effect several times the resampling effect, and equally with none that six operators can establish.

More rollouts lower power at a fixed ratio, from 18% to 13% at a ratio of two (Fig. 2). Because s_A more than halved, a ratio of two now means a partition SD of only 0.063, 1.40 times the Arm B residual rather than 2.78 times; the residual, which includes partition-by-seed interaction, shrank less. Simulation error is about 2.7 points per estimate.

D. Tier labels

At 50 rollouts the Square partitions looked like a tier effect: holding out either “better” operator gave 0.493 and 0.453, against at most 0.68 for the others, a between-tier share of 0.94 against a null mean of 0.40. Even then it was a two-operator pattern: exactly 3 of the 15 ways of pairing the six partitions reach the observed share, precisely those keeping the two “better” operators together (permutation $p = 0.200$), and the tier pairing ranks third among them. At 500 rollouts it is gone: the between-tier share is 0.43, 7 of 15 pairings reach it, within-tier differences (0.075) match between-tier ones (0.071), and the first “better” partition moved from 0.493 to 0.624. Amendment 3 withdrew our Stage 1 statement that the secondary prediction was contradicted; we make no tier claim in either direction. A pattern across six three-seed, 50-rollout means was strong enough to be written down and did not survive ten times the rollouts.

TABLE II
SEED EFFECT AND RESOLVABILITY FROM ARM A (n STATES, TEN SEEDS). FLOORS AS IN SECTION III-C; COCHRAN’S p TESTS FOR ANY DIFFERENCE BETWEEN SEEDS ON SHARED STATES. LAST TWO COLUMNS: SMALLEST DIFFERENCE BETWEEN TWO THREE-SEED MEANS DISTINGUISHABLE AT 95% (EQ. 1) ON SHARED (PAIRED) OR INDEPENDENT (UNPAIRED) STATES, t QUANTILE IN PARENTHESES. ALL BUT THE SEED SD ARE POST HOC (AMENDMENT 4).

Task	n	seed SD	floor		Cochran p	min. resolvable diff. (points)	
			binomial	per-state		paired	unpaired
Lift	50	0.017	0.015	0.015	0.213	2.7 (3.8)	2.8 (4.0)
Can	50	0.023	0.032	0.032	0.855	3.7 (5.3)	5.1 (7.2)
Square	50	0.079	0.069	0.064	0.133	12.7 (18.0)	13.4 (18.9)
Square	500	0.031	0.021	0.020	0.009	5.0 (7.1)	5.2 (7.3)

E. What the convention can resolve

Two methods reported as means of k seeds differ detectably at a two-sided 95% level only if their true difference exceeds roughly

$$\Delta_{\min} \approx 1.96 s \sqrt{2/k}, \quad s^2 \approx \sigma_c^2 + v/n, \quad (1)$$

where σ_c^2 is the variance of checkpoints’ true success rates across seeds. If both methods use the same stored states (paired), $v = q_i(1 - q_i)$ and s is simply s_A ; if each draws its own states (unpaired), as with unseeded resets, $v = p(1 - p)$, which is never smaller. For $k = 3$ (Table II), Square at 50 rollouts needs 12.7 points paired (18.0 with a t quantile) and 13.4 unpaired, and 5.0 paired at 500 rollouts. The gap is largest on Can, 3.7 against 5.1, where many states are solved by every checkpoint. These thresholds cover seed and rollout noise only, not demonstration resampling (Arm C partition SD 0.014 on Square at 500 rollouts) or any operator effect.

F. Exploratory analyses

Declared exploratory in Amendment 2, not tests of the preregistered claims: (i) with the binomial floor removed from its denominator, R is 2.20 on Square at 50 rollouts and 2.36 at 500, but the correction is undefined in 32% and 10% of resamples; (ii) the exact F -test for a partition component gives $p = 0.003$ in Arm B against 0.121 in Arm C on Square at 50 rollouts, and 0.012 against 0.349 at 500, with no test below 0.05 on Lift or Can; (iii) partitions holding out a “better” operator are 17.2 points below the rest at 50 rollouts and 7.5 at 500. These are consistent with a partition effect in Arm B on Square and change no preregistered decision.

V. DISCUSSION AND RECOMMENDATIONS

The hypothesis we registered was about operators; the clearest finding is about rollouts. With 50 rollouts the evaluation cannot see differences between seeds on any of the three tasks, so a three-seed standard deviation describes the measurement more than the method, and on Square the point estimate is a weak proxy for the same checkpoint’s success rate under a larger evaluation. The motivating statement, that comparisons a few points apart are not resolvable from the reported statistic, holds on Square for a more elementary reason than the one we proposed. The operator question remains open: both stages order the Square point estimates the same way, neither

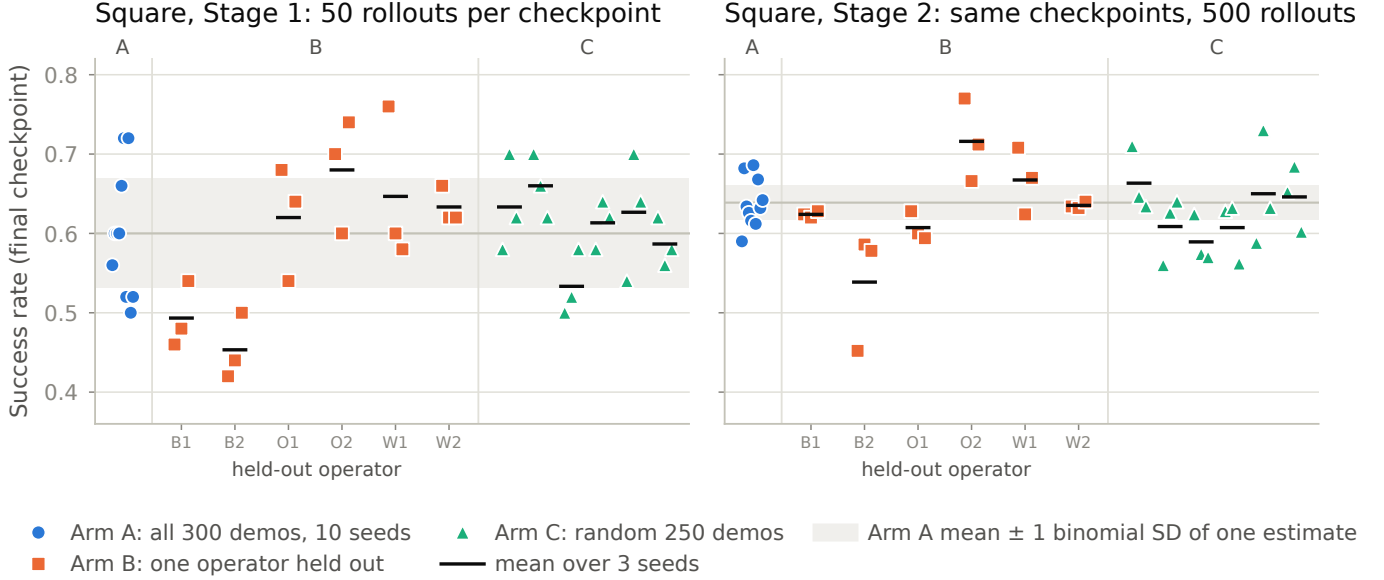


Fig. 3. Square: the same 46 final checkpoints evaluated with 50 rollouts (Stage 1, left) and with 500 rollouts on 500 new fixed initial conditions (Stage 2, right); layout as in Fig. 1. The low means of the two “better”-operator partitions at 50 rollouts do not persist at 500.

establishes the effect, and at 500 rollouts power at a ratio of two is still 13%. A dataset with more operators is required. We recommend:

- 1) **Report the rollout noise next to the seed SD, and test the seed effect on shared states.** A seed SD near $\sqrt{p(1-p)/n}$ means the evaluation may not see the seed effect; with stored states, Cochran’s Q on per-rollout outcomes is an exact test of whether checkpoints differ at all.
- 2) **Use enough rollouts for the success band.** Near 60% success, 50 rollouts give a standard error of 7 points per checkpoint and 500 give 2; Eq. (1) states what a budget can resolve.
- 3) **Fix and publish the initial conditions and per-rollout outcomes,** including the scene model where it is randomised, on a set large enough that conclusions are not conditional on it.
- 4) **Evaluate a checkpoint chosen without the evaluation rollouts, and state the training-set composition** when using subsets of a multi-operator dataset, without reading structure from a handful of low-rollout partition means.

VI. LIMITATIONS

Six operators per task is the binding limitation and sets the power of every operator-level test. The study uses one architecture, low-dimensional observations, one training length, simulation only, one benchmark family, and three seeds per partition. Partitions within an arm overlap heavily, so the random-effects model is an approximation; Arm C shares that structure, which is why the contrast with it is the meaningful quantity. Stage 2 was designed after Stage 1 and reuses its checkpoints. The exact enumeration was added after the Stage 2 contrast was seen at the boundary, and the preregistered computation alone would have supported that contrast.

The per-state floor, Cochran’s test, unpaired thresholds and explicit decision rule were added after review and change no Stage 1 decision.

VII. CONCLUSION

We preregistered the claim that operator composition contributes more variance than training seed on the robomimic Multi-Human benchmarks, ran 138 training runs with a matched random control, and did not confirm it; the experiment was underpowered for that claim by the number of operators, not by compute. It was well powered for a different conclusion: with 50 rollouts the conventional three-seed statistic cannot detect differences between seeds on any task, cannot resolve differences between methods smaller than about 12.7 points on Square, and produced a tier pattern that a tenfold larger evaluation did not reproduce. The preregistration and amendments, partitions, stored initial conditions, per-rollout results for both stages (6,900 and 23,000 rollouts), and the scripts that regenerate every number, table and figure here are archived at <https://doi.org/10.5281/zenodo.23109850>.

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